AA 273 Midterm

The midterm can be taken at any time during the 24 hour period it is available but must be completed within **a single 5 hour period**. The exam must be submitted no later than 3:00 PM (Pacific) on Saturday, February 7th through Gradescope.

Allowed resources

- Canvas (e.g., homework solutions, lecture notes/videos, starter code, past Ed posts)
- Textbooks
- Built-In Coding Assistants (ie Cursor, Copilot)
- Online documentation for any libraries that have been included in starter code (homework and exam). Examples include Numpy, Scipy, Matplotlib, Open3D, Pytorch, Pandas and standard Python libraries (math, csv).

Resources that are **NOT allowed**

- Material from previous iterations of the class (e.g., homework or exams)
- Discussion with classmates or anyone outside of the class
- The internet (except for documentation for the aforementioned libraries)
- Generative AI tools (except for code editing)
- You are not allowed to post questions relating to the midterm publicly on Ed between May 5 and May 7.

Additional Notes

- We prefer typeset solutions. However, if you choose to handwrite your solutions, the answers must be legible.
- Some questions require coding. Please provide associated code in your submission.
- If there are any questions regarding the exam, please direct them to the CAs through Ed as a private message or through email.

Q1 (20 points)

Controlled Bayes Filter

A discrete state $X_t \in \{1, 2, 3\}$ evolves according to a controlled transition model that depends on a control input $u_t \in \{\mathrm{N}, \mathrm{F}\}$ (No-move vs Forward). The "forward" direction is $1 \to 2 \to 3 \to 1$.

Controlled transition model.

- If $u_t = \mathrm{N}$: $p(x_t = x_{t-1}) = 0.9$, and the remaining 0.1 probability is split equally among the other two states.
- If $u_t = \mathrm{F}$: with probability 0.8 you go forward (i.e., $x_t = x_{t-1} + 1 \bmod 3$), and with probability 0.2 you stay (i.e., $x_t = x_{t-1}$).

Observation model. The observation is $Y_t \in \{1, 2, 3\}$ with

$$p(Y_t = y \mid X_t = x) = \begin{cases} 0.7, & y = x, \\ 0.15, & y \neq x \text{ (each of the other two values).} \end{cases}$$

Initial belief:

$$b_0 = [0.2,\ 0.5,\ 0.3]$$

corresponding to states $[1, 2, 3]$.

Controls and observations:

$$u_1 = \mathrm{F},\ y_1 = 2; \qquad u_2 = \mathrm{N},\ y_2 = 2.$$

(a) **[6 Points]** Write the Bayes filter recursion for

$$b_t(x) = p(X_t = x \mid y_{1:t}, u_{1:t}),$$

including a prediction step using $p(x_t \mid x_{t-1}, u_t)$ and an update step using $p(y_t \mid x_t)$.

Solution: Define the predicted belief

$$\bar{b}_t(x_t) := p(x_t \mid y_{1:t-1}, u_{1:t}).$$

By the law of total probability and the controlled Markov property,

$$\boxed{\bar{b}_t(x_t) = \sum_{x_{t-1}} p(x_t \mid x_{t-1}, u_t)\, b_{t-1}(x_{t-1})}$$

Then by Bayes' rule and conditional independence of y_t given x_t,

$$b_t(x_t) \propto p(y_t \mid x_t)\, \bar{b}_t(x_t), \qquad \boxed{b_t(x_t) = \eta\, p(y_t \mid x_t)\, \bar{b}_t(x_t)}$$

with normalization

$$\eta^{-1} = \sum_{x_t} p(y_t \mid x_t)\, \bar{b}_t(x_t).$$

(b) **[10 Points]** Compute b_1 and b_2 numerically.

Solution: First write the transition matrices.

For $u = \mathrm{F}$ (forward with prob 0.8, stay with prob 0.2), with rows x_{t-1} and columns x_t:

$$T^{\mathrm{F}} = \begin{bmatrix} 0.2 & 0.8 & 0 \\ 0 & 0.2 & 0.8 \\ 0.8 & 0 & 0.2 \end{bmatrix}.$$

For $u = \mathrm{N}$ (stay with prob 0.9, otherwise split 0.05 and 0.05):

$$T^{\mathrm{N}} = \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.9 & 0.05 \\ 0.05 & 0.05 & 0.9 \end{bmatrix}.$$

For $y = 2$, the likelihoods are

$$p(y = 2 \mid x) = [0.15,\ 0.7,\ 0.15] \text{ for } x = 1, 2, 3.$$

Time 1: $u_1 = \mathrm{F}$, $y_1 = 2$.

Prediction:

$$\bar{b}_1 = b_0\, T^{\mathrm{F}} = [0.2,\ 0.5,\ 0.3] \begin{bmatrix} 0.2 & 0.8 & 0 \\ 0 & 0.2 & 0.8 \\ 0.8 & 0 & 0.2 \end{bmatrix} = [0.28,\ 0.26,\ 0.46].$$

Update (unnormalized):

$$\tilde{b}_1 = \bar{b}_1 \odot [0.15,\ 0.7,\ 0.15] = [0.042,\ 0.182,\ 0.069],$$

normalizer $c_1 = 0.042 + 0.182 + 0.069 = 0.293$, hence

$$\boxed{b_1 = \frac{1}{0.293}[0.042,\ 0.182,\ 0.069] \approx [0.1433,\ 0.6212,\ 0.2355].}$$

Time 2: $u_2 = \mathrm{N}$, $y_2 = 2$.

Prediction:

$$\bar{b}_2 = b_1\, T^{\mathrm{N}} \approx [0.1433,\ 0.6212,\ 0.2355] \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.9 & 0.05 \\ 0.05 & 0.05 & 0.9 \end{bmatrix} \approx [0.171843,\ 0.577986,\ 0.250171].$$

Update (unnormalized):

$$\tilde{b}_2 = \bar{b}_2 \odot [0.15,\ 0.7,\ 0.15] \approx [0.025776,\ 0.404590,\ 0.037526],$$

normalizer $c_2 \approx 0.467892$, hence

$$\boxed{b_2 \approx \frac{1}{0.467892}[0.025776,\ 0.404590,\ 0.037526] \approx [0.05509,\ 0.86471,\ 0.08020].}$$

(c) **[4 Points]** Let $u_3 = \mathrm{F}$. Compute the predicted belief

$$\bar{b}_3(x) = p(X_3 = x \mid y_{1:2}, u_{1:3})$$

(i.e., prediction only; no measurement update at time 3).

Solution: Prediction only:

$$\bar{b}_3 = b_2\, T^{\mathrm{F}} \approx [0.05509,\ 0.86471,\ 0.08020] \begin{bmatrix} 0.2 & 0.8 & 0 \\ 0 & 0.2 & 0.8 \\ 0.8 & 0 & 0.2 \end{bmatrix} \approx [0.07518,\ 0.21701,\ 0.70781].$$

$$\boxed{\bar{b}_3 \approx [0.07518,\ 0.21701,\ 0.70781].}$$

enumerate

Q2 (20 points)

Quadratic Expectations and the Trace Trick

Let $X \in \mathbb{R}^n$ be a random vector with mean $\mu = \mathbb{E}[X]$ and covariance $\Sigma = \mathbb{E}[(X-\mu)(X-\mu)^\mathsf{T}]$. Let $Q \in \mathbb{R}^{n\times n}$ be a **symmetric** matrix, i.e. $Q = Q^\mathsf{T}$, and let $b \in \mathbb{R}^n$ be a constant vector.

Hint: For any vectors u, v and matrix A of compatible sizes,

$$u^\mathsf{T} A v = \mathrm{tr}\big(A v u^\mathsf{T}\big)\,.$$

(a) **[10 Points]** Derive a closed-form expression for $\mathbb{E}[\|QX+b\|_2^2]$ in terms of (μ, Σ, Q, b). Your final answer should separate a "mean" term and a "covariance" term. **Hint:** $\Sigma = \mathbb{E}[XX^\mathsf{T}] - \mu\mu^\mathsf{T}$.

Solution: Expand the squared norm:

$$\|QX+b\|_2^2 = (QX+b)^\mathsf{T}(QX+b) = X^\mathsf{T}Q^\mathsf{T}QX + 2b^\mathsf{T}QX + b^\mathsf{T}b.$$

Since Q is symmetric, $Q^\mathsf{T} = Q$, so $Q^\mathsf{T}Q = Q^2$. Taking expectation,

$$\mathbb{E}[\|QX+b\|_2^2] = \mathbb{E}[X^\mathsf{T}Q^2X] + 2b^\mathsf{T}Q\,\mathbb{E}[X] + b^\mathsf{T}b = \mathbb{E}[X^\mathsf{T}Q^2X] + 2b^\mathsf{T}Q\mu + b^\mathsf{T}b.$$

Use the trace trick: $X^\mathsf{T}AX = \mathrm{tr}(AXX^\mathsf{T})$:

$$\mathbb{E}[X^\mathsf{T}Q^2X] = \mathbb{E}[\mathrm{tr}(Q^2XX^\mathsf{T})] = \mathrm{tr}\big(Q^2\,\mathbb{E}[XX^\mathsf{T}]\big)\,.$$

Using $\mathbb{E}[XX^\mathsf{T}] = \Sigma + \mu\mu^\mathsf{T}$ (derivation in (d) below if included),

$$\mathbb{E}[X^\mathsf{T}Q^2X] = \mathrm{tr}(Q^2\Sigma) + \mathrm{tr}(Q^2\mu\mu^\mathsf{T}) = \mathrm{tr}(Q^2\Sigma) + \mu^\mathsf{T}Q^2\mu.$$

Therefore,

$$\mathbb{E}[\|QX+b\|_2^2] = \mathrm{tr}(Q^2\Sigma) + \mu^\mathsf{T}Q^2\mu + 2b^\mathsf{T}Q\mu + b^\mathsf{T}b.$$

Recognize $\mu^\mathsf{T}Q^2\mu + 2b^\mathsf{T}Q\mu + b^\mathsf{T}b = \|Q\mu+b\|_2^2$, so

$$\boxed{\mathbb{E}[\|QX+b\|_2^2] = \|Q\mu+b\|_2^2 + \mathrm{tr}(Q^2\Sigma).}$$

here the first term is the "mean" term and the second is the "variance"

(b) **[7 Points]** Using the hint above, show that

$$\mathbb{E}\big[(X-\mu)^\mathsf{T}Q(X-\mu)\big] = \mathrm{tr}(Q\Sigma).$$

Solution: Let $\delta := X - \mu$. Then $\mathbb{E}[\delta\delta^\mathsf{T}] = \Sigma$. Using the trace trick,

$$\delta^\mathsf{T}Q\delta = \mathrm{tr}(Q\delta\delta^\mathsf{T}).$$

Taking expectation and using linearity of trace and expectation,

$$\mathbb{E}[\delta^\mathsf{T}Q\delta] = \mathbb{E}[\mathrm{tr}(Q\delta\delta^\mathsf{T})] = \mathrm{tr}\big(Q\,\mathbb{E}[\delta\delta^\mathsf{T}]\big) = \mathrm{tr}(Q\Sigma).$$

$$\boxed{\mathbb{E}[(X-\mu)^\mathsf{T}Q(X-\mu)] = \mathrm{tr}(Q\Sigma).}$$

(c) **[3 Points]** If in addition $Q \succeq 0$ (Q is PSD), give a one-line argument why $\mathbb{E}[(X-\mu)^\mathsf{T}Q(X-\mu)] \geq 0$.

Solution: If $Q \succeq 0$, then $v^\mathsf{T}Qv \geq 0$ for all v. Taking $v = X - \mu$ gives $(X-\mu)^\mathsf{T}Q(X-\mu) \geq 0$. Taking expectation preserves inequality so:

$$\boxed{Q \succeq 0 \;\Rightarrow\; \mathbb{E}[(X-\mu)^\mathsf{T}Q(X-\mu)] \geq 0.}$$

Q3 (20 points)

Sensor Fusion and Error Probability

Consider an autonomous vehicle in $\mathbb{R}^2$. Assume the prior over the robot pose state X is Gaussian:

$$X \sim \mathcal{N}(\mu_0, \Sigma_0).$$

At some arbitrary time we are given two measurements Y_1 and Y_2 that are conditionally independent given X, with Gaussian measurement models:

$$(Y_1 \mid X = x) \sim \mathcal{N}(x, \Sigma_1), \qquad (Y_2 \mid X = x) \sim \mathcal{N}(x, \Sigma_2).$$

(a) **[5 Points]** You will fuse these two measurements. Using Bayes' rule, first compute the fused posterior $p(x \mid y_1, y_2) = \mathcal{N}(\mu_f, \Sigma_f)$, **and then** state how the magnitude of Σ_f compares to that of the individual measurement covariances. You may work with the unnormalized posterior distribution and ignore constants that do not depend on x, because the Σ_f which you will compute normalizes the Gaussian via the normalization constant η. g **Solution:**

$$p(x \mid y_1, y_2) \propto, p(y_1, y_2, \mid x)\, p(x) \qquad \text{(Bayes' Rule)}$$

$$p(x \mid y_1, y_2) \propto p(y_1 \mid x)\, p(y_2 \mid x)\, p(x) \qquad \text{(Cond. Indep.)}$$

Ignoring constants (w.r.t. x), write each term in exponential form:

$$p(x) \propto \exp\Big(-\tfrac{1}{2}(x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0)\Big)$$

$$p(y_1 \mid x) \propto \exp\Big(-\tfrac{1}{2}(y_1-x)^\top \Sigma_1^{-1}(y_1-x)\Big)$$

$$p(y_2 \mid x) \propto \exp\Big(-\tfrac{1}{2}(y_2-x)^\top \Sigma_2^{-1}(y_2-x)\Big)$$

Multiply and collect quadratic and linear terms in x, then match to a Gaussian in canonical form. The posterior is:

$$\boxed{p(x \mid y_1, y_2) = \mathcal{N}(\mu_f, \Sigma_f)}$$

with

$$\boxed{\begin{aligned} \Sigma_f^{-1} &= \Sigma_0^{-1} + \Sigma_1^{-1} + \Sigma_2^{-1} \\ \mu_f &= \Sigma_f \left(\Sigma_0^{-1}\mu_0 + \Sigma_1^{-1} y_1 + \Sigma_2^{-1} y_2\right) \end{aligned}}$$

Comparison: since Σ_f^{-1} adds positive semidefinite information terms, Σ_f is **smaller (in the PSD sense)** than each covariance used alone (e.g., smaller than the posterior covariance using only one measurement).

(b) **[5 Points]** Now let's make our system more efficient by replacing measurements Y_1 and Y_2 with a single sensor that outputs a scalar linear Gaussian measurement W with measurement model

$$W = C^\top X + V$$

where $C \in \mathbb{R}^n$ is a vector of constants and $V \in \mathbb{R}$ is noise, independent of X, and distributed as

$$V \sim \mathcal{N}(0, R).$$

Find $\mathbb{E}[W]$ and $\mathrm{Cov}(W)$, and identify the distribution of W.

Solution:

Because W is an affine function of a Gaussian random variable, it is Gaussian.

$$\boxed{\begin{aligned} \mathbb{E}[W] &= \mathbb{E}[C^\top X + V] \\ &= C^\top \mathbb{E}[X] + \mathbb{E}[V] \\ &= C^\top \mu_0 \end{aligned}}$$

$$\boxed{\begin{aligned}\mathrm{Cov}(W) &= \mathrm{Var}(C^\top X + V) \\ &= \mathrm{Var}(C^\top X) + \mathrm{Var}(V) \\ &= C^\top \Sigma_0 C + R\end{aligned}}$$

$$\boxed{W \sim \mathcal{N}\big(C^\top \mu_0,\ C^\top \Sigma_0 C + R\big)}$$

(c) **[5 Points]** We would like to know the probability that the measurement W lies within ε of the expected value of that measurement. Find

$$\mathbb{P}(|W - \mathbb{E}[W]| \le \varepsilon)\,.$$

Your answer should be in terms of the standard normal cumulative distribution function (CDF), defined for a scalar standard normal random variable Z as $\Phi(a) = Pr(Z \le a) = \int_{-\infty}^{a} \mathcal{N}(0,1)\,dz$.
Hint: $\mathbb{P}(a \le Z \le b) = \Phi(b) - \Phi(a)$ for $Z \sim \mathcal{N}(0,1)$.

Solution:

From part (b),

$$W \sim \mathcal{N}(\mu_W, \Sigma_W), \qquad \mu_W = C^\top \mu_0, \quad \Sigma_W = C^\top \Sigma_0 C + R.$$

The event is

$$|W - \mu_W| \le \varepsilon \quad \Longleftrightarrow \quad -\varepsilon \le W - \mu_w \le \varepsilon.$$

Standardize with

$$Z \;=\; \frac{W - \mu_W}{\sigma_W} \;\sim\; \mathcal{N}(0,1).$$

Then

$$\boxed{\begin{aligned}\mathbb{P}(|W - \mathbb{E}[W]| \le \varepsilon) &= \mathbb{P}(-\varepsilon \le W - \mu_W \le \varepsilon) \\ &= \mathbb{P}\left(-\frac{\varepsilon}{\sigma_W} \le Z \le \frac{\varepsilon}{\sigma_W}\right) \\ &= \Phi\left(\frac{\varepsilon}{\sigma_W}\right) - \Phi\left(-\frac{\varepsilon}{\sigma_W}\right) \\ &= 2\,\Phi\left(\frac{\varepsilon}{\sigma_W}\right) - 1\end{aligned}}$$

(d) **[5 Points]** Now suppose we swap the sensor and the new measurement model for W is:

$$W = \cos(X).$$

You are thinking about applying a Kalman filter to the task of estimating the mean and covariance for W over time. Will the KF work? Why or why not?

Solution:

A standard (linear) Kalman filter will not directly work because it assumes a **linear** measurement model of the form $y = Cx + v$ and maintains an exactly Gaussian posterior under linear-Gaussian assumptions.

Here $W = \cos(X)$ is nonlinear, so even if X is Gaussian, the induced distribution of W is not Gaussian in general, and the posterior $p(x \mid w)$ is not guaranteed to be Gaussian.

To use Kalman-style filtering, you would typically apply a **nonlinear** variant such as an **Extended Kalman Filter (EKF)** (linearize $\cos(\cdot)$ about the current estimate) or an **Unscented Kalman Filter (UKF)** (sigma-point approximation).

Q4 (15 points)

Kalman Filter with Feedback Controller

Previously, we assumed a vehicle's control input u_t is deterministic, however in many systems control is state-dependent. Assume U_t is a random variable that is determined through a proportional state feedback policy as

$$U_t = K_p X_t$$

where K_p is the deterministic proportional gain matrix.

Assume the following dynamics and measurement model:

$$X_{t+1} = AX_t + BU_t + W_t$$

$$Y_t = CX_t + V_t$$

where W_t and V_t are zero mean GWN with covariances Q and R, respectively. The noise processes are independent of each other and the initial state X_0.

(a) **[2 Points]** Will the standard Kalman Filter applied to the system matrices A, B, C work for this system? Why or why not?

Solution: No, because some steps in the derivation of the Kalman filter require u_t to be deterministic.

(b) **[3 Points]** Derive new system matrices A_{cl}, B_{cl}, and C_{cl} to yield a new linear-Gaussian state space model with U_t eliminated (i.e., find the closed loop system model). Can we use the Kalman filter for this model? Why or why not?

Solution:

$$X_{t+1} = AX_t + B(K_p X_t) + W_t$$

$$X_{t+1} = (A + K_p B)X_t + W_t$$

$$\boxed{\begin{aligned} A_{cl} &= A + K_p B \\ B_{cl} &= 0 \\ C_{cl} &= C \end{aligned}}$$

Yes, because the system only depends on X_t and not U_t, we can use the KF on this model.

(c) **[5 Points]** Suppose you apply the Kalman filter to find $\mu_{t|t}$, $\Sigma_{t|t}$. Using these values, find expressions for the expected value and covariance of the posterior of U_t, given measurement history $Y_{1:t}$.

Solution:

Expected Value:

$$\mathbb{E}[U_t \mid Y_{1:t}]$$

Substitute expression for U_t:

$$\mathbb{E}[K_p X_t \mid Y_{1:t}]$$

K_p is a constant, take it out:

$$K_p \mathbb{E}[X_t \mid Y_{1:t}]$$

We find:

$$\boxed{\mathbb{E}[U_t \mid Y_{1:t}] = K_p \mu_{t|t}}$$

Covariance:

$$\text{Cov}(U_t \mid Y_{1:t})$$

Substitute expression for U_t:

$$\text{Cov}(K_p X_t \mid Y_{1:t}) = E[(K_p X_t - K_p \mu_{t|t})(K_p X_t - K_p \mu_{t|t})^T \mid Y_{1:y}]$$

Pull constant K_p out on the right and left:

$$= K_p E[(X_t - \mu_{t|t})(X_t - \mu_{t|t})^T \mid Y_{1:t}] K_p^T$$

We find:

$$\boxed{\text{Cov}(U_t \mid Y_{1:t}) = K_p \Sigma_{t|t} K_p^T}$$

(d) **[5 Points]** Now suppose your control policy is stochastic, meaning the control input is given by:

$$U_t = K_p X_t + N_t$$

with noise $N_t \sim GWN(0, \Sigma_n)$, independent of X_0 and the W_t and V_t noise processes. Show that you can form a modified linear-Gaussian state space model with A_{cl}, B_{cl}, C_{cl} as above, but with a modified Q_{cl}, for a new process noise covariance. Find Q_{cl}. Assuming you run the KF with this new noise covariance to get $\mu_{t|t}$ and $\Sigma_{t|t}$ find new expressions for $E[U_t \mid Y_{1:t}]$ and $Cov(U_t \mid Y_{1:t})$ for this case.

Solution:

$$X_{t+1} = AX_t + B(K_p X_t + N_t) + W_t$$

$$X_{t+1} = (A + K_p B)X_t + (BN_t + W_t)$$

$$X_{t+1} = A_{cl} X_t + W_t'$$

We find Q_{cl} as follows

$$\begin{aligned} Q_{cl} &= Cov(W_t') = E[(W_t + BN_t)(W_t + BN_T)^T] \\ &= E[W_t W_t^T] + E[W_t N_t^T] B^T + BE[N_t W_t^T] + BE[N_t N_t^T] B^T \\ &= Q + B\Sigma_n B^T \end{aligned}$$

so W_t' is GWN, distributed by $W_t' \sim \mathcal{N}(0, Q')$, where:

$$\boxed{Q' = Q + B\Sigma_n B^T}$$

Similarly to part (c) above, we find

$$\boxed{E[U_t \mid Y_{1:t}] = K_p \mu_{t|t} + E[N_t] = K_p \mu_{t|t}}$$

and

$$\boxed{Cov(U_t \mid Y_{1:t}) = K_p \Sigma K_p^T + \Sigma_n}$$

Q5 (25 points + 5 points extra credit)

Save Mark Watney!

You are an engineer at NASA working on the rescue of Mark Watney. Watney is planning to use his spacesuit to propel himself towards commander Lewis, who will catch him. Because of this risky maneuver, you are tasked with estimating Watney's position with respect to Commander Lewis.

The simplified dynamics are given by:

$$x_{t+1} = x_t + \Delta t\, \dot{x}_t$$

$$\dot{x}_{t+1} = \dot{x}_t + \Delta t\, \ddot{x}_t + W_t$$

where $x_t \in \mathbb{R}^3$, $\dot{x}_t \in \mathbb{R}^3$, $\ddot{x}_t \in \mathbb{R}^3$, are Watney's position, velocity and acceleration, respectively [1]. Watney is also wearing an MPS (Martian Positioning System) receiver, which broadcasts his position with respect to Commander Lewis, via the measurement model:

$$y_t = x_t + V_t$$

With noises $W_t \sim \mathcal{N}(0, Q)$ and $V_t \sim \mathcal{N}(0, R)$, which are uncorrelated with each other and the initial state.



Figure 1: Rescue of Mark Watney, *from The Martian, 2015*

(a) **[3 Points]** What is the state vector and the system matrices A, B, C?

Solution:

Let's write this in discrete state-space form:

$$x = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}_{6\times 1} = \begin{bmatrix} I_{3\times 3} & \Delta t\, I_{3\times 3} \\ 0_{3\times 3} & I_{3\times 3} \end{bmatrix}_{6\times 6} \begin{bmatrix} x_t \\ \dot{x}_t \end{bmatrix}_{6\times 1} + \begin{bmatrix} 0_{3\times 3} \\ \Delta t\, I_{3\times 3} \end{bmatrix}_{6\times 3} u_t + \begin{bmatrix} 0_{3\times 3} \\ I_{3\times 3} \end{bmatrix}_{6\times 3} w_t$$

$$y = \begin{bmatrix} y_t \end{bmatrix}_{3\times 1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}_{3\times 6} \begin{bmatrix} x_t \\ \dot{x}_t \end{bmatrix}_{6\times 1} + \begin{bmatrix} I_{3\times 1} \end{bmatrix}_{3\times 1} v_t$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} \quad A = \begin{bmatrix} I_{3\times 3} & \Delta t\, I_{3\times 3} \\ 0_{3\times 3} & I_{3\times 3} \end{bmatrix}_{6\times 6} \quad B = \begin{bmatrix} 0_{3\times 3} \\ \Delta t\, I_{3\times 3} \end{bmatrix}_{6\times 3} \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}_{3\times 6}$$

[1] If you were solving this problem in real life, you would use the Hill-Clohessy-Wiltshire Equations, which are used to estimate the relative position and velocity between two objects in a circular orbit

(b) **[2 Points]** Watney will start at Position $x_0 = [406,\ -203,\ -4.06]$ m, and Velocity $\dot{x}_0 = [0,\ 0,\ 0]$ m/s, and will fly with a constant acceleration $u_t = [-1.0,\ 0.5,\ 0.01]$ m/s^2. Simulate and attach a plot of Watney's trajectory, along with measurements from MPS, to ensure he will rendez-vous within 1m of Commander Lewis at $x_L = [0,\ 0,\ 0]$ after 30 seconds of flight. Assume $\Delta t = 1$ sec, and there is no noise, meaning $Q = 0$, $R = 0$.

Hint 1: You are encouraged to use your code from PS3.

Hint 2: If you are not converging to $x_f = [0, 0, 0]$ within 1 meter, ensure you are using $\Delta t = 1$ sec.

Solution:

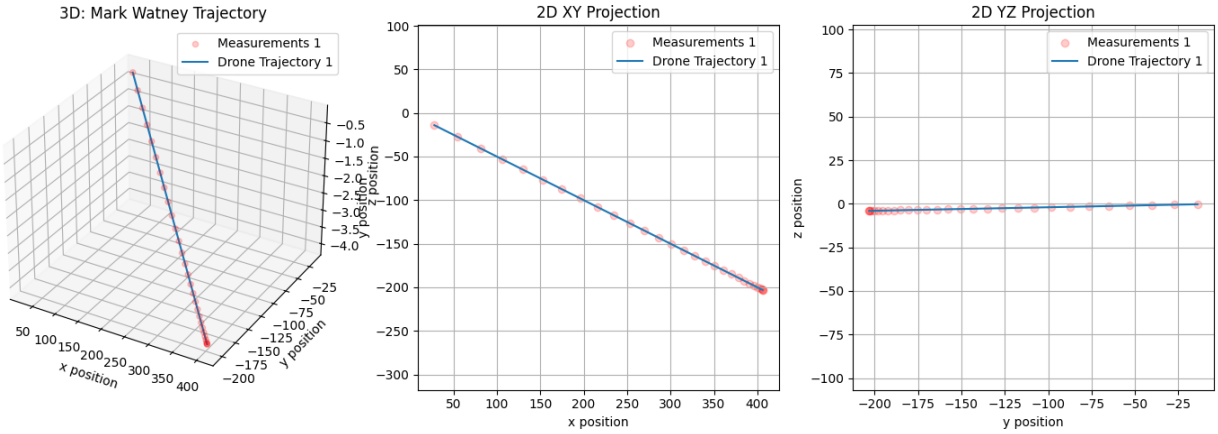


Figure 2: Solution to Question 2

(c) **[2 Points]** You speak up and suggest that Watney's trajectory is not perfect since there is noise! Through field testing you compute $Q = 0.1 \cdot I_3$ m^2/s^2 and $R = 100 \cdot I_3$ m^2. Now you simulate Watney's trajectory - does he meet Commander Lewis at the end of his trajectory?

Note: For consistency, please use `np.random.seed(273)`.

Solution:

No, his final position is:

$$x_f = [23.3,\ -39.5,\ 2.9] \text{ m}$$

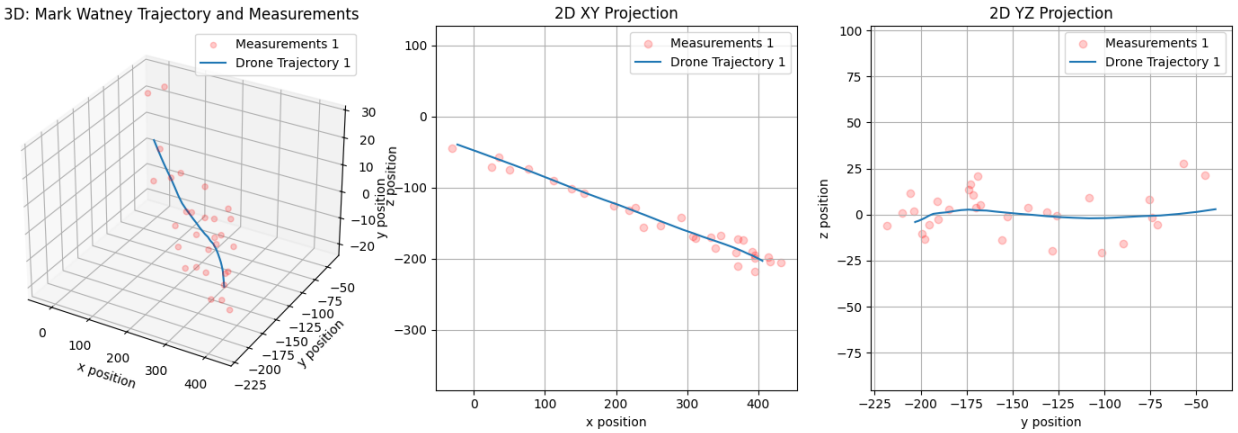


Figure 3: Solution to Question 3

(d) **[3 Points]** Nervous, your boss tasks you with implementing a Kalman Filter to estimate Watney's position. Again, we think Watney is starting from

$$x_0 = (406,\ -203,\ -4.06,\ 0,\ 0,\ 0)$$

but we are not sure so we have an initial covariance of

$$P_0 = \begin{bmatrix} 10^3 I_3 \text{ m}^2 & 0 \\ 0 & I_3 \text{ m}^2/\text{s}^2 \end{bmatrix}$$

Attach a plot of the noisy ground truth trajectory, the measurements, and your estimated position.

Note: Your noisy dynamics are exclusively in terms of velocity, so you will need to modify your Q matrix.

Solution:

The resulting trajectory is exactly identical. Applying this transformation has no impact.

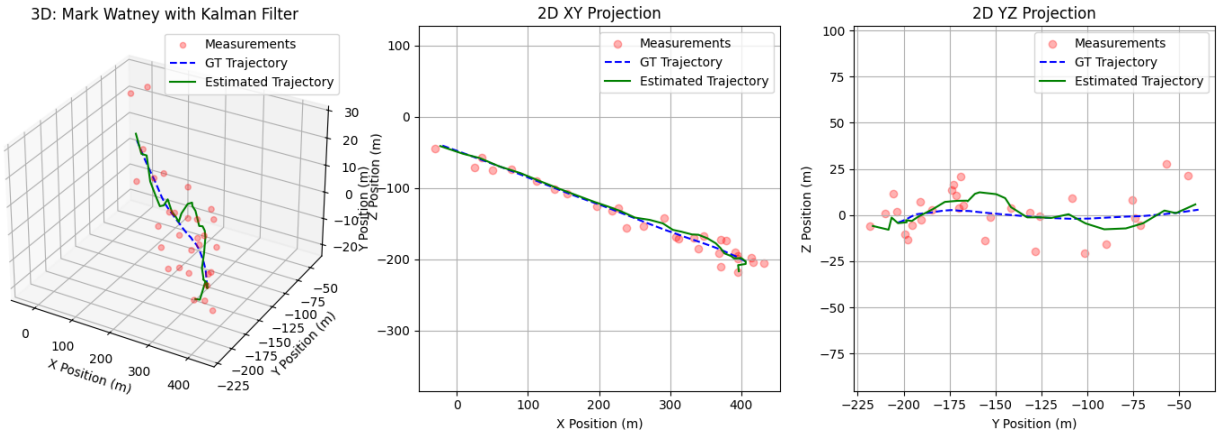


Figure 4: Solution to Question 4

(e) **[2 Points]** In addition to not rendezvousing with Lewis, the estimated trajectory has poor performance. In order to fix the Kalman Filter performance, your boss suggests multiplying both the Q, R matrices and initial covariance P_0 by 10. Using your simulation explain why this is not helpful for changing the mean state estimate.

Solution:

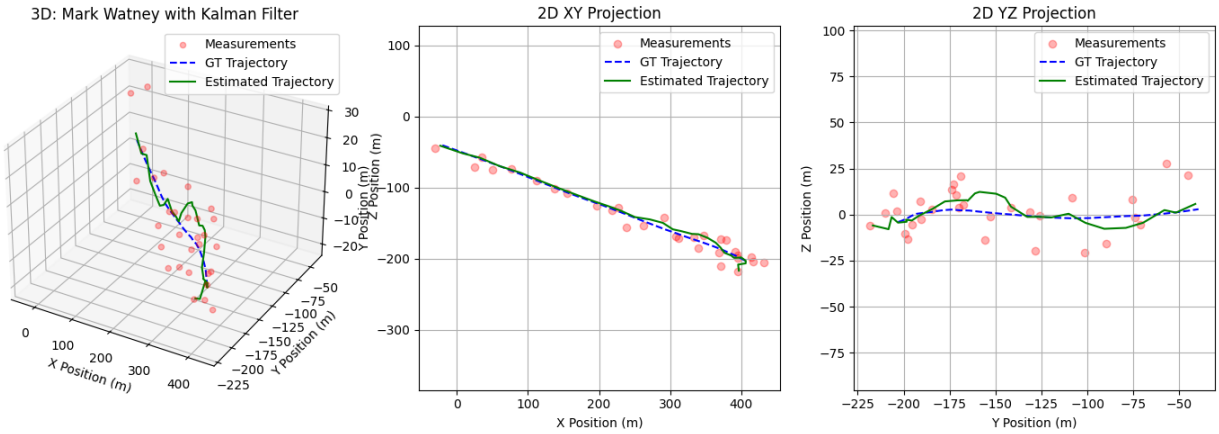


Figure 5: Solution to Question 5

(f) **[6 Points]** Surprised by your result, you decide to prove that if Q, R and P_0 are scaled by the same constant λ, it has the same impact on the mean state estimate as not scaling Q, R, P_0 at all. What is the effect on state covariance?

Solution:

Unscaled case:

$$\hat{x}'_{t+1} = A\hat{x}_t + Bu_t$$

$$P'_{t+1} = AP_tA^T + Q$$

$$K = P'_{t+1}C^T\left(CP'_{t+1}C^T + R\right)^{-1}$$

$$\hat{x}_{t+1} = \hat{x}'_{t+1} + K\left(y_t - C\hat{x}'_{t+1}\right)$$

$$P_{t+1} = (I - KC)\, P'_{t+1}$$

Scaled case:

$$Q = \lambda Q, \qquad R = \lambda R, \qquad P'_{t+1} = \lambda P'_{t+1}$$

$$\Rightarrow \hat{x}'_{t+1} = A\hat{x}_t + Bu_t \quad \text{(unchanged)}$$

$$\Rightarrow P'_{t+1} = A\left(\lambda P_t\right)A^T + \lambda Q$$

$$= \lambda\left(AP_tA^T\right) + \lambda Q$$

$$= \lambda\left[AP_tA^T + Q\right]$$

$$= \lambda\left[P'_{t+1}\right]$$

$$\text{so} \quad P'_{t+1} = \lambda P'_{t+1}$$

$$\Rightarrow K = P'_{t+1}C^T\left(CP'_{t+1}C^T + \lambda R\right)^{-1}$$

$$= \lambda P'_{t+1}C^T\left(C\lambda P'_{t+1}C^T + \lambda R\right)^{-1}$$

$$= \lambda P'_{t+1}C^T\left(\lambda\left[CP'_{t+1}C^T + R\right]\right)^{-1}$$

$$= \lambda P'_{t+1}C^T\frac{1}{\lambda}\left(CP'_{t+1}C^T + R\right)^{-1}$$

$$= P'_{t+1}C^T\left(CP'_{t+1}C^T + R\right)^{-1}$$

$$\Rightarrow K = K \ \checkmark$$

we see that Kalman gain is unchanged
Therefore the output state estimate is also unchanged:

$$\hat{x}_{t+1} = \hat{x}'_{t+1} + K\left(y_t - C\hat{x}'_{t+1}\right)$$

However the output covariance is larger!

$$P_{t+1} = (I - KC)\, P'_{t+1}$$

$$= (I - KC)\,\lambda P'_{t+1}$$

$$= \lambda\left[(I - KC)\, P'_{t+1}\right]$$

$$\Rightarrow P_{t+1} = \lambda P_{t+1} \ \checkmark$$

The output covariance is larger by a factor of λ!
Note: This can also be proved via induction. Just showing the graphs is not enough.

(g) **[2 Points]** In order to have better filter performance, you claim you need better measurements. Your boss says he can move the MPS satellites for better coverage, and can give you measurements with either $R = 1 \cdot I_3$ or $R = 0.1 \cdot I_3$. Which is better? Defend your answer with simulation results, and briefly explain what is happening in the filter.

Solution:

We can clearly see via simulation that a smaller R value (0.1) has better filter performance. Since noise is filtered from R, a smaller noise variance results in a smoother tracked trajectory. Notice that the dead-reckoned trajectory is unchanged!

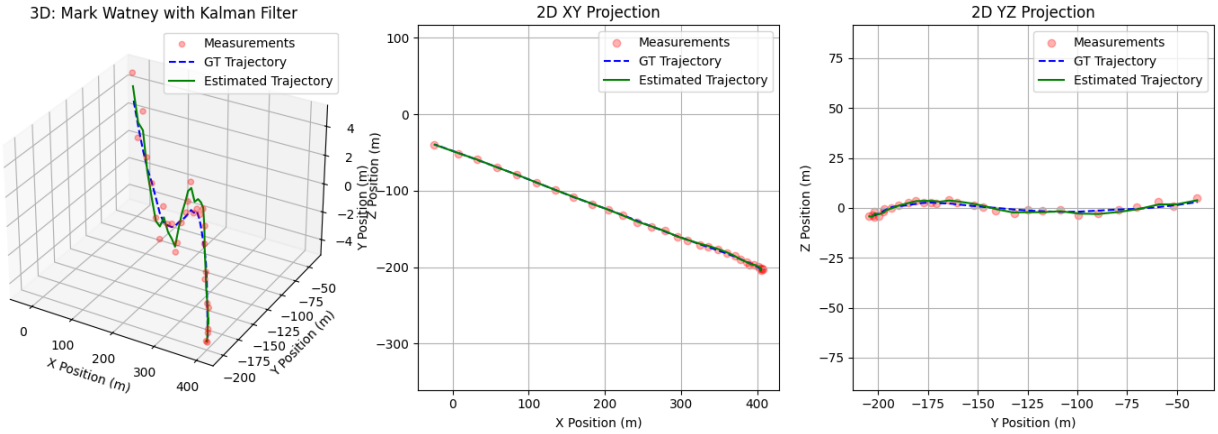


Figure 6: Result with R scaled by 1

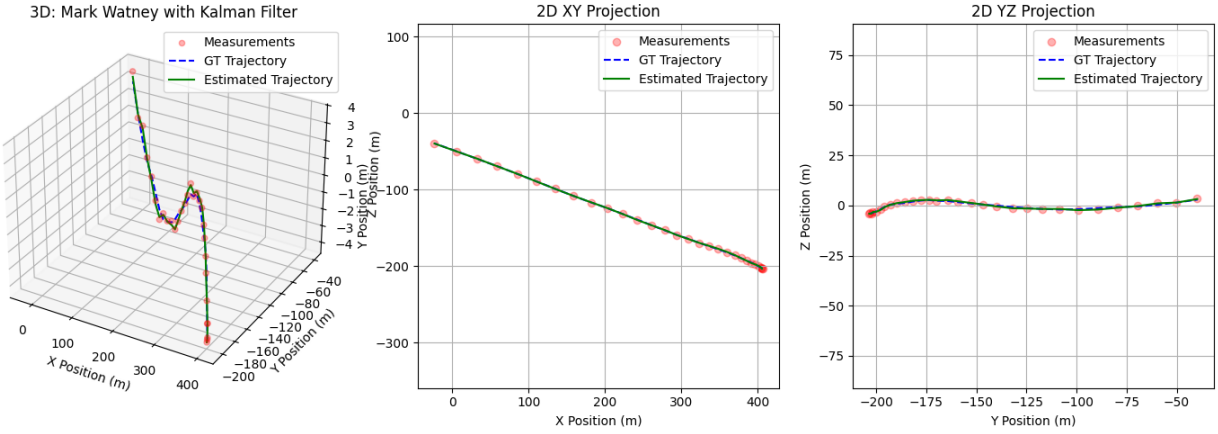


Figure 7: Result with R scaled by 0.1

(h) **[3 Points]** Your boss is satisfied with the filter performance, but unhappy that Mark is not rendezvousing with Commander Lewis. By only changing Q, R, P_0, how can you get Mark's trajectory to end within 1 meter of Commander Lewis' position.

Solution: The only way for the trajectory to converge without control is to set $\mathbf{Q} \approx 0$. By testing numbers, if you set $\mathbf{Q} \approx 10^{-3}\mathbf{I_3}$, the trajectory will converge to within 1 meter. His final position is:

$$x_f = [-0.23,\ -0.39,\ 0.03] \text{ m}$$

(i) **[2 Points]** In reality the dynamics function is given by the relative 2-body problem:

$$x_{t+1} = x_t + \Delta t\, \dot{x}_t + \frac{1}{2}\Delta t^2\, \ddot{x}_t$$

$$\dot{x}_{t+1} = \dot{x}_t + \Delta t\, \ddot{x}_t$$

$$\ddot{x}_{t+1} = -\mu \left(\frac{r_L + x_t}{\|r_L + x_t\|^3} - \frac{r_L}{\|r_L\|^3} \right)$$

Where μ is the Martian gravitational constant, and r_L is the time varying position vector between the center of Mars and Lewis. Your boss says this should be compatible with your existing Kalman Filter, though you claim it violates a critical assumption. What assumption does this violate?

Solution: These equations are clearly non-linear, so this violates the linearity principle of the Kalman Filter.

(j) **Extra Credit 1 [2 Points]** While these dynamics cannot be used with the standard Kalman Filter, they are solvable with a different mtehod. What method is this?

Solution: The Extended Kalman Filter (EKF)

(k) **Extra Credit 2 [3 points]** What is the physical interpretation to your solution of part (h). Do you think this is tractable in this problem? If not, what would you do to get Watney's trajectory to converge with Commander Lewis?

Solution: A near-zero **Q** means there is virtually no uncertainty in Watney's ability to control his own dynamics (obviously not true), and that the dynamics equations are accurate (also not true, we know these are simplified and even with the 2-body problem dynamics these do not account for perturbation forces). In reality to actually converge the trajectory, control actions must be taken (ie PID) to converge the trajectory.